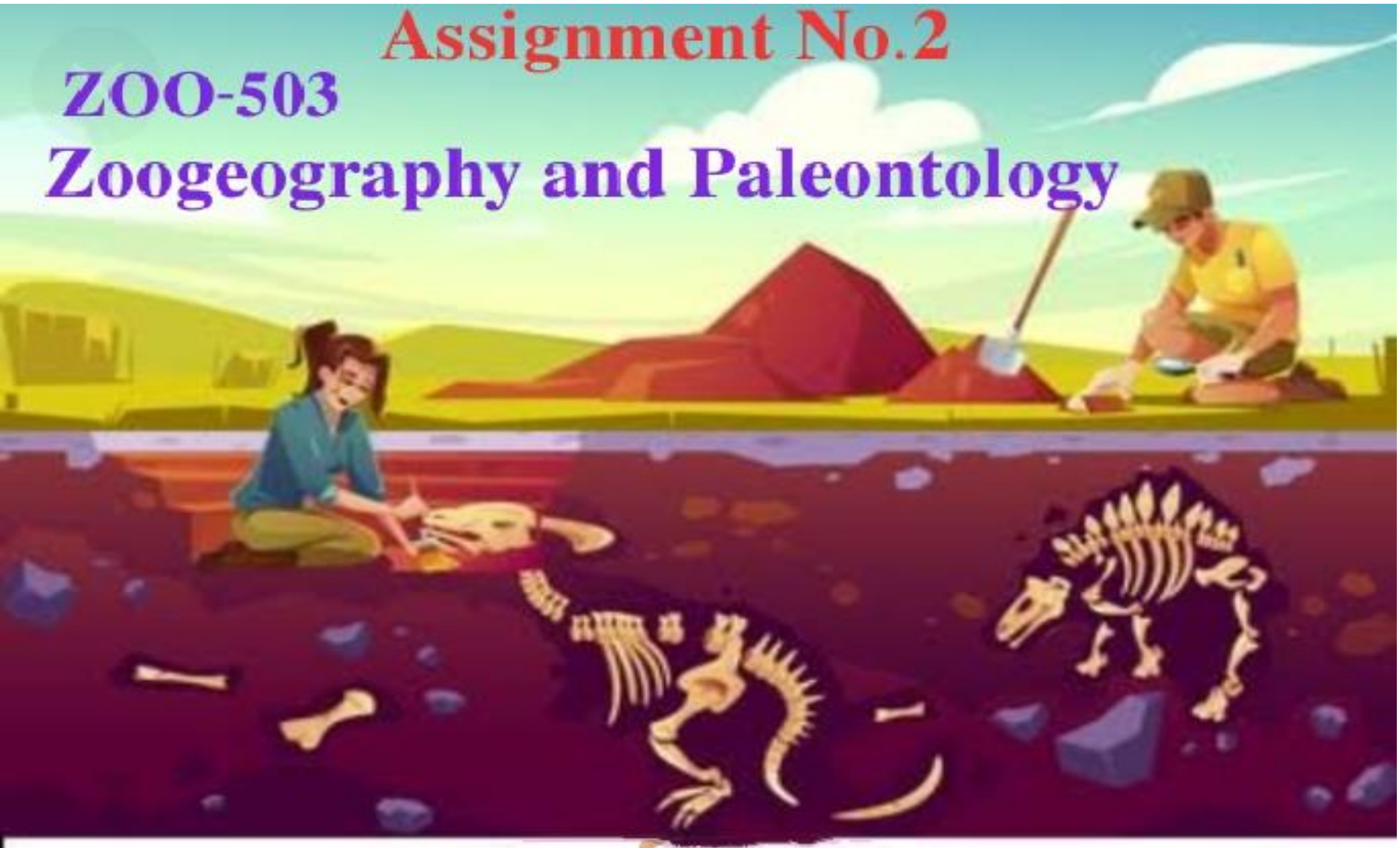
Assignment No.2

ZOO-503

Zoogeography and Paleontology



By

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[# AQUARIUS #]

Fossil Plants:

Introduction:

Plant fossils are the remains of very old plants, preserved in rocks. Plant fossils can be imprints, compressions, casts, molds, or petrified fossils. Plant fossils (macroscopic and microscopic remains) provide direct evidence of vegetation throughout Earth's history. In some cases, plant fossils are found in growth position. In-place (autochthonous) fossils in wetland facies include buried stumps of wetland trees (Fig. 1I), and coal balls (permineralized peat) in coal beds. Coal balls can preserve cellular details of the peat-forming flora. In many cases, plant fossils represent transported (allochthonous) elements. Some fossil plants (megascopic and macroscopic) have

extant members that are hydrophytic; hence, interpretation of wetland habitats or ecologies for these taxa is straightforward. Interpretations for other taxa are based on comparisons to related extant plants, on their own morphological features, or on their associations with wetland facies.

Example:

Petrified Wood, Leaf Imprints & Plants In Amber. Plants that flourished on earth millions of years ago are perfectly preserved as fossils. In petrified wood, the tissue of ancient trees is completely replaced by minerals, converting trunks and branches into stone.

Formation:

Water and layering of sediment is usually necessary for fossils to form, so fossils are often formed in rivers, lakes or estuaries, and they can be found in areas that were rivers, lakes or estuaries in the past. The oldest plant fossils are about 500 million years old and were found in Argentina.

Compression fossils are often formed in water where heavy sediment flattens leaves or other plant parts. … Fossil molds and casts are formed when animal or plant tissues become surrounded by hardened sediment the tissue then decays. The hollow negative created by the tissue is called a mold.

Preservation:

Plant fossils are commonly preserved in fine-grained sediment such as sand, silt, or clay, or in association with organic deposits such as peat (coal). As you look at the various modes of preservation in lab, note the characteristics of the rock matrix in which plant material is preserved.

Importance:

Paleobotany is the study of plants that lived long ago. It is one half of a branch of science called paleontology that studies how life has evolved on Earth over millions of years. Paleobotanists study fossilized plant life to get information about the types of plants that lived during different time periods.

Paleobotany studies help us to understand the past vegetation, climate and palaeoecology including the history of plant evolution, biostratigraphy, the existence of Coal/Petroleum can be known from the study of paleobotany because they are formed from dead remains of living organisms(fossils).

Classification of Fossil Plants:

Classification on the basis of Body Parts:

Plant fossils are classified into two groups:

1.Macrofossils

These are the larger plant members, such as stems and leaves, usually detached and often fragmented.

2. Microfossils

These are the shells or resistant outer walls of minute plants, such as diatoms, or of pollen grains and spores.

Classification on the basis of Formation:

1.Compression Fossils

These plant fossils are two-dimensional imprints of plants that have been transformed during the fossilization process and smashed into a flat impression of the plant they once were. These fossils often retain some organic matter. Common examples of compression fossils can be found in coal and peat, which contain many types of accumulated fossil plants.

2.Impression Fossils

Impression fossils are similar to compression fossils in that they are both two-dimensional, but these fossils are not remnants of the plant itself and do not contain organic material. Impression fossils essentially leave an imprint of the plant material in some fine-grained or soft sediment, such as clay or silt. Once the plant matter decays, the impression remains to be fossilized.

3.Cast and Mold Fossils

Mold and cast fossils are three-dimensional fossils that sometimes retain some organic material. These fossils are formed when sediment fills in an empty space in the plant (casts) or by surrounding the plant itself before the plant decays (molds). These fossils tend to record outside features of the plant but do not reveal cellular information.

4.Permineralization Fossils

Permineralization is a process through which the plant material becomes impregnated or permeated by mineral rich liquid before the plant matter decays. The minerals seep into the plant and then harden, forming the three-dimensional fossil. Because the liquid can seep into all parts of the plant, these fossils often yield highly detailed information about the plant's internal structure. Petrified wood is a common example of a permineralization fossil.

5.Compaction Fossils

Like compression, compaction fossils are reduced volume versions of the plants, though they are three-dimensional and are not generally mineralized. These fossils retain organic material and are often found in peat, lignite coals and soft sediments.

6.Molecular Fossils

Molecular fossils are the chemical remains of the plant material. Molecular fossils are those that reveal the chemical make-up of the plant and can even contain fossilized DNA and RNA. Mass spectrometry, spectrophotometry and other advanced chromatographic techniques are often employed when studying molecular fossil materials.